

THREE RIVERS, MI, USA

WMO: 720371

Lat: 41.960N

Lon: 85.593W

Elev: 824

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 54850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	5.1	-8.0	3.7	0.6	-2.0	5.1	6.4	24.3	27.3	20.8	26.1	4.9	210	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.7	72.5	86.9	71.3	84.2	70.0	75.9	84.7	74.4	82.5	73.0	80.5	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	127.1	79.9	72.1	122.3	78.8	70.5	115.8	77.2	39.9	84.7	38.4	82.5	37.1	80.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	17.8	16.0	-9.4	93.4	5.6	3.4	-13.5	95.9	-16.8	97.9	-19.9	99.8	-24.0	102.3		
			-9.9	78.5	5.5	1.9	-13.9	79.8	-17.1	80.9	-20.2	81.9	-24.2	83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	24.3	26.0	37.5	48.3	60.6	69.1	72.2	70.5	64.3	52.5	39.8	31.1
	DBStd	19.07	11.25	11.12	11.17	9.03	8.77	6.03	5.78	5.22	7.50	8.75	8.82	9.08
	HDD50	3026	796	675	416	136	14	0	0	0	1	75	325	588
	HDD65	6266	1260	1093	855	504	192	28	9	14	104	400	756	1051
	CDD50	2954	1	1	28	86	342	573	688	634	429	151	19	2
	CDD65	719	0	0	2	3	55	151	231	184	82	11	0	0
	CDH74	6231	0	0	19	45	583	1234	2128	1433	691	97	1	0
	CDH80	1815	0	0	1	3	149	353	724	369	201	15	0	0
(14) Wind	WSAvg	6.3	7.7	7.5	7.4	8.2	6.5	5.5	4.3	4.0	4.5	6.0	6.7	7.3
(15) Precipitation	PrecAvg	39.4	2.7	2.1	2.5	3.2	4.5	3.9	4.0	4.2	3.5	3.4	3.0	2.5
	PrecMax	47.3	5.7	4.6	4.8	5.8	8.6	7.6	8.9	12.8	11.7	7.4	5.7	4.7
	PrecMin	32.2	1.0	0.8	0.8	0.4	1.6	1.7	1.5	0.7	1.1	0.7	1.1	0.7
	PrecStd	4.7	1.4	1.1	1.0	1.3	1.9	1.7	1.8	2.4	2.3	1.8	1.3	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	60.9	76.6	79.3	88.3	90.5	95.6	90.3	90.3	82.5	70.0	59.8
		MCWB	53.1	53.8	64.5	60.9	67.7	71.5	73.4	72.5	72.1	67.5	58.0	56.2
	2%	DB	49.1	53.4	67.6	73.2	83.9	87.6	90.2	86.8	86.2	77.0	63.5	53.3
		MCWB	46.6	49.9	58.3	58.5	66.2	70.1	74.1	71.7	72.0	63.9	55.4	49.7
	5%	DB	43.1	45.7	60.5	69.6	80.9	84.1	87.2	84.1	81.5	72.2	58.5	47.7
		MCWB	40.3	42.2	52.6	56.5	65.4	69.4	72.3	70.8	68.7	62.4	51.9	44.9
	10%	DB	38.8	40.4	54.5	64.1	76.8	81.1	83.9	81.6	77.7	67.7	53.7	43.1
		MCWB	36.7	37.1	48.2	53.6	63.7	67.9	70.5	69.3	66.4	60.3	48.5	40.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	55.3	65.2	65.2	72.0	76.1	78.4	76.9	75.5	71.1	61.5	57.4
		MCDB	55.5	57.6	76.3	74.0	81.1	85.6	88.1	84.9	85.8	79.4	64.9	59.8
	2%	WB	47.3	49.0	59.9	61.6	69.8	74.1	76.6	75.1	73.5	67.7	56.6	50.0
		MCDB	48.4	53.2	65.7	69.6	78.7	82.4	85.7	82.4	82.3	72.5	62.2	52.2
	5%	WB	41.1	42.5	54.7	58.4	67.9	72.3	74.8	73.7	71.5	64.2	52.5	44.7
		MCDB	43.1	45.3	59.0	66.2	76.4	80.1	83.6	80.3	78.0	68.7	57.1	46.8
	10%	WB	36.8	37.2	48.8	55.4	65.7	70.5	73.0	72.0	69.4	61.0	48.5	40.6
		MCDB	38.7	39.9	54.1	63.9	74.1	77.4	80.8	78.3	75.1	67.1	53.2	43.0
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.6	17.0	19.9	20.5	18.7	20.0	19.5	20.6	18.2	14.7	11.5
		MCDBR	15.5	19.6	24.8	26.7	25.4	21.8	22.3	22.0	23.4	24.0	22.0	17.7
	5% WB	MCWBR	13.5	16.2	16.5	14.9	11.8	9.2	8.6	9.5	10.5	13.2	15.1	15.2
		MCWBR	14.9	19.3	22.0	22.8	21.2	17.7	18.9	16.9	18.7	19.4	20.2	16.4
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.346	0.352	0.393	0.426	0.439	0.435	0.441	0.400	0.354	0.319	0.301
		taud	2.362	2.220	2.336	2.243	2.198	2.202	2.220	2.197	2.291	2.419	2.464	2.409
	Ebn at Noon	Ebn at Noon	268	268	283	278	270	266	266	260	263	264	259	257
		Edh at Noon	26	36	36	42	45	45	44	43	37	29	24	23
(44) All-Sky Solar Radiation	RadAvg	RadAvg	447	712	1110	1470	1695	1918	1947	1686	1369	880	545	373
	RadStd	RadStd	43	101	107	134	164	125	94	91	119	91	79	44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page